

1 Question (20 points)

Give state diagrams of NFA's with the specified number of states recognizing each of the following languages:

- $0^*1^*0^+$ (3 states)
- w contains the substring 0101 (5 states)

2 Question (20 points)

Let $B_{(m,n)} = \{x \mid x \text{ is a number in modulus } m \text{ that is a multiple of } n\}$. Show that for each $m, n \geq 1$, the language is $B_{(m,n)}$ is regular.

3 Question (20 points)

Give CFG's with the specified number of variable recognizing each of the following languages:

- $A = \{w \mid \text{the length of } w \text{ is odd and its middle symbol is a } 1\}$ (1 variable)
- $B = \{w \mid \text{the set of strings over alphabet } \{a, b\} \text{ with more } a\text{'s than } b\text{'s}\}$ (2 variables)

4 Question (20 points)

Design a single tape TM that recognizes the language $L = \{0^n 1^{n^2} | n \geq 1\}$. Do not describe the steps of TM. Draw a state diagram, give states, transitions, etc.

5 Question (20 points)

Let a k -PDA be a pushdown automaton that has k stacks. Thus a 0-PDA is an NFA, 1-PDA is conventional PDA. Prove that 2-PDA's are as powerful as TM's by simulating a Turing machine tape with two stacks.